2019B F=ma Exam: Problem 2

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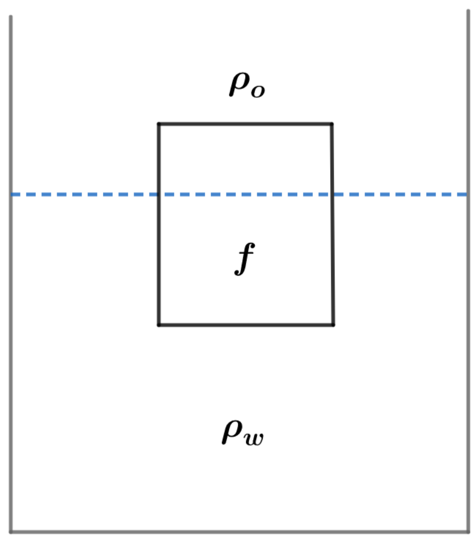

If a fraction f of an object is submerged when it is placed in water, we have

$$F_g = F_B$$
$$\rho_o V g = \rho_w f V g$$
$$f = \frac{\rho_o}{\rho_w}$$

For the plastic block, we have

$$f = \frac{1}{2} = \frac{\rho_{plas}}{\rho_w}$$
$$\rho_{plas} = \frac{\rho_w}{2}$$

The effective density of the wood and plastic combination is $\rho_1 = \frac{\rho_{wood}+\rho_{plas}}{2}$ so

$$f = \frac{1}{2} = \frac{\rho_1}{\rho_w}$$
$$\rho_{wood} + \rho_{plas} = \rho_w$$
$$\rho_{wood} = \frac{\rho_w}{2}$$

The effective density of the styrofoam and plastic combination is $\rho_2 = \frac{\rho_{sty}+\rho_{plas}}{2}$ so

$$f = \frac{1}{2} = \frac{\rho_2}{\rho_{oil}}$$

$$\rho_{sty} + \rho_{plas} = \rho_{oil}$$

$$\rho_{sty} + 0.5\rho_w = 0.7\rho_w$$

$$\rho_{sty} = 0.2\rho_w$$

Hence,

$$\frac{\rho_{wood}}{\rho_{sty}} = 2.5$$

so the answer is $\boxed{\text{A}}$.